

Overshoot



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Abstract Overshoot is the state in which human demand, within a given time period, exceeds the amount ecosystems regenerate in the same time period. Overshoot is possible because the accumulated stock of ecosystems can be depleted. The accumulation of greenhouse gases in the atmosphere is one of the most prominent symptoms of overshoot. But global depletion is not a long-term option. Therefore, global overshoot will inevitably end.

Overshoot is the state in which human demand, within a given time period, exceeds the amount ecosystems regenerate in the same time period (Meadows et al., 1972a; Catton, 1980). It means that demand is larger than what ecosystems regenerate. Humanity has become a depleting force of the biosphere due to global overshoot, and is now a major force shaping the planet's condition. Hence the size of global overshoot is a core parameter of the →*Anthropocene*.

Overshoot is possible because ecosystems' accumulated stocks can be depleted. Examples of overshoot include cutting trees faster than the forest can renew them, or overgrazing pastures. Symptoms of current global overshoot include deforestation, overfishing, groundwater depletion, soil loss, and accumulation of greenhouse gases in the atmosphere. As global depletion is not a long-term option given the limited size of accumulated stocks that can be depleted and limited capacities of waste sinks that can be filled, global overshoot will inevitably end. The question is only whether it ends by design or disaster.

Given that most environmental ills, such as climate change, biodiversity loss, deforestation, freshwater scarcity, and desertification, are ultimately driven by humanity's excess use of the biosphere, it is surprising that overshoot is barely recognized in the public discussion as the key dynamic driving ecological depletion. For instance, the media service Meltwater (2023) finds that throughout all of 2022,

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only 476 news articles on the web mentioned “ecological overshoot”. One application of the concept, →Earth Overshoot Day, was referenced in 10 times as many articles: 4770 during the same period. “SDGs” or “sustainable development goals” which are just one, potential insufficient (Wackernagel et al. 2017), strategy to counteract overshoot, was found in 1.46 million articles in that year. “Climate change”, which is merely one symptom of overshoot, appeared in 6.43 million articles. This demonstrates that the phenomenon of ecological overshoot as a key driver of environmental trends is vastly underestimated and underappreciated in academia, policy circles, and among the public.

Given its implications for resource security, conflict, and economic stability, ecological overshoot may be one of the most dominant risks to human wellbeing in the twenty-first century. This risk is only surpassed by the self-imposed threat of ignoring it, thereby staying underprepared.

Overshoot, or the difference between what ecosystems can provide and the amount of resources being used, can be measured in different ways: →*Net Primary Productivity* (biomass accumulation over a year) is compared to HANPP or human appropriation of net primary productivity (Vitousek et al., 1986; Haberl et al., 2007), but it is challenging to define a sustainable level of appropriation; →*carrying capacity*, or the number of animal units that can be sustained by a habitat, is compared to actual populations; or →*biocapacity* (regeneration measured in area which is scaled proportionally to its potential net primary productivity) compared to →*ecological footprints*. →*Planetary boundaries* are closely related to overshoot, but measure components of it, not the whole (Steffen et al., 2015). Planetary boundaries indicate conditions that, if breached, would lead to global overshoot. For each of the nine identified dimensions, thresholds are identified which, if exceeded, can lead to the triggering of irreversible shifts in ecosystems.

The dynamics of overshoot have been modelled by various researchers, including in early “predator and prey models” by Alfred Lotka and Vito Volterra in the 1920s (Brauer & Castillo-Chavez, 2000). A prominent computer-based study of global overshoot and its long-term implications was “Limits to Growth” published in 1972 (Meadows et al., 1972b). Others have modelled carrying capacity as a logistical function that asymptotically reaches carrying capacity. However, in the real world, demand can exceed regeneration if feedbacks are weak or delayed, leading to large overshoot. For instance, the feedback of climate change on emissions comes with large time delays, thereby not benefiting from self-corrective forces. In contrast, if feedbacks are strong and relatively fast, even light overshoot might cause some oscillations in the amounts ecosystems can provide to its hosts. If ecological feedback is weak or time delayed, the most likely scenario of overshoot is a gradual (as with forests) or rapid (as with fisheries) reduction of the amount that hosts can harvest. The extent of such a crash following unmanaged overshoot depends on how depleted stocks already are and how much such depletion reduces regeneration rates.

Global overshoot can be observed for carbon emissions alone. For instance, the Intergovernmental Panel on Climate Change (IPCC)’s 2014 report states that a concentration of greenhouse gases in the atmosphere of 450 ppm CO₂ equivalent gives

humanity a 66% chance to comply with the Paris Agreement's 2-degree Celsius (2 °C) goal (IPCC, 2014). 450 ppm of a gas means that there is an occurrence of 450 molecules of this gas per million molecules of air (ppm stands for parts per million). CO₂ equivalent is a measure of all relevant greenhouse gases (minus water vapour) expressed in the amount of CO₂ that would produce the equivalent amount of warming over a 100-year period. In contrast to those 450 ppm CO₂equivalent, the National Oceanic and Atmospheric Administration of the United States Department of Commerce (or NOAA) reports that in 2020, our planet's atmosphere already reached a greenhouse gas concentration of 504 ppm CO₂ equivalent (Butler & Montzka, 2021). This indicates that the greenhouse gas dimension alone is already in severe overshoot. This also illustrates the relationship between deficit spending (annual emissions) and the accumulated debt (greenhouse gas concentration in the atmosphere), run somewhat parallel to financial deficit and debt. As with finances, low levels of ecological debt are tolerable, but as ecological deficit spending continues, the debt becomes so large that its impact becomes difficult to manage. For instance, increasing the carbon debt by augmenting greenhouse gas concentrations from 300 to 305 ppm may not have significant consequences, but adding 5 ppm at the current level of greenhouse gas concentrations radically increases the likelihood of runaway climate change (Steffen et al., 2015; Randers & Goluke, 2020).

Taking all competing demands on regeneration into account makes the picture even clearer. Based on UN statistics, adding up those competing demands for regeneration and comparing them to what the planet's ecosystems can regenerate, the National Footprint and Biocapacity Accounts conclude that by 2018, human demand exceeded planetary regeneration by 75% (York University et al., 2022). These *→ecological footprint*-based accounts also estimate the ratio for each country. Also, for countries, it is possible to run a Footprint that is larger than the country's biocapacity. Three mechanisms allow for such a country deficit:

1. Net-importing biocapacity from elsewhere (this means that the ecological footprint embodied in imports exceeds the ecological footprint embodied in exports);
2. Using the global commons (such as in the case of fishing international waters or emitting greenhouse into the global atmosphere); and
3. Overusing or overshooting one's own territorial biocapacity (overharvesting forests or fish stocks, etc.). This would be local overshoot: use of the local ecosystems beyond their ability to regenerate.

At the global level, the ecological deficit is identical to overshoot because the globe as a whole does not trade, and cannot use an extraterritorial "global commons". Therefore, the difference between humanity's ecological footprint and the planet's biocapacity is both the global deficit as well as global overshoot.

Overshoot is maintained by liquidating stocks of ecological resources and accumulating waste, such as greenhouse gases in the atmosphere. Continuous deficit spending accumulates in an ever-larger debt, and one that may not be easily reversible.

To increase the prominence and recognition of overshoot risks, Global Footprint Network runs an annual campaign called “→*Earth Overshoot Day*”. From January 1 to that day, humanity has on aggregate used as much from the biosphere as the biosphere can regenerate in the entire year. According to the media search facility [Meltwater.com](https://meltwater.com), Earth Overshoot Day generated over 7 billion media impressions in 2022.

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Dr. Mathis Wackernagel created the *footprint* concept in the early 1990s with Prof. William E. Rees. The carbon footprint has become the most popular variant. In 2003, he founded Global Footprint Network, a sustainability think-tank, making planetary constraints relevant to decision-making. Its largest engagement campaign is its annual Earth Overshoot Day. Mathis's honors include the 2018 World Sustainability Award, the 2015 IAIA Global Environment Award, and the 2012 Blue Planet Prize.

Dr. David Lin leads Global Footprint Network's research team, and contributes to the production, development, and improvement of the National Footprint and Biocapacity Accounts. Prior to joining Global Footprint Network, David earned his Ph.D. and worked as a post-doctoral researcher in the Systems Ecology Laboratory at the University of Texas at El Paso. His research focused on integrating models of ecosystem function with land cover change analysis in Arctic ecosystems. David is a native of California, and holds a BS in ecology, behavior, and evolution from the University of California, Los Angeles.